

There are various ways in software to handle error conditions. In C or Go, one returns error code. Other programming languages like C++ or Java prefer to throw *exceptions*. One benefit of using exceptions is that it keeps your code mostly clean since the error-handling code is often separate.

It is debatable whether handling exceptions is better than dealing with error codes. I will happily use one or the other.

What I will object to, however, is the use of exceptions for control flow. It is fine to throw an exception when a file cannot be opened, unexpectedly. But you should not use exceptions to branch on the type of a value.

Let me illustrate.

Suppose that my code expects integers to be always positive. I might then have a function that checks such a condition:

```
int get_positive_value(int x) {  
    if(x < 0) { throw std::runtime_error("it is not positive!"); }  
    return x;  
}
```

So far, so good. I am assuming that the exception is normally never thrown. It gets thrown if I have some kind of error.

If I want to sum the absolute values of the integers contained in an array, the following branching code is fine:

```
int sum = 0;  
for (int x : a) {  
    if(x < 0) {  
        sum += -x;  
    } else {  
        sum += x;  
    }  
}
```

Unfortunately, I often see solutions abusing exceptions:

```
int sum = 0;
for (int x : a) {
    try {
        sum += get_positive_value(x);
    } catch (...) {
        sum += -x;
    }
}
```

The latter is obviously ugly and hard-to-maintain code. But what is more, it can be highly inefficient. To illustrate, I wrote a small benchmark over random arrays containing a few thousand elements. I use the LLVM clang 12 compiler on a skylake processor. The normal code is 10000 times faster in my tests!

normal code	0.05 ns/value
exception	500 ns/value

Your results will differ but it is generally the case that using exceptions for control flow leads to suboptimal performance. And it is ugly too!

Daniel Lemire, "Avoid exception throwing in performance-sensitive code," in *Daniel Lemire's blog*, May 13, 2022,
<https://lemire.me/blog/2022/05/13/avoid-exception-throwing-in-performance-sensitive-code/>.
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PUBLISHED BY



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